Batched CM Switches and Certified Theta Periods

A Scoped Ram Calculus Study of the Rational Hodge Claim for CM Abelian Varieties

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Abstract

The September 30 source manuscript claims the rational Hodge conjecture for all CM abelian varieties, using four-factor CM switches, theta differentials on projective ball quotients, and arithmetic extraction of rational CM sources. We map that complete argument but do not certify its global geometric and arithmetic inputs. Our main local improvement batches independent embedding-coordinate corrections into a single two-row switch. For p rows and s conjugate embedding pairs, at most $\min\{s\lfloor p/2\rfloor, \binom{p}{2}\}$ switch invocations suffice, with an exact matching formulation for the optimal monotone batching. The corresponding algebraic correspondence construction retains every auxiliary factor, polarization contraction, and sign. A second local result makes the source's sparse theta-gradient limit quantitative through an explicit lattice-shell tail and finite wedge margin. Rational orbit projectors, mixed-period error budgets, and exact rational-span tests are then organized into a conditional certificate protocol. None constructs a missing surface, proves a numerical projector algebraic, or infers cycle existence from approximate periods. Contribution assessment: C3/5, the embedding-independent switch bound and explicit sparse-theta certificate are notable local improvements, without a new global Hodge theorem.

1 Scope, status, and the proof architecture

Let $A/\mathbb{C}$ be an abelian variety of dimension g. Write

$$\mathrm{Hdg}^{2p}(A) = H^{2p}(A,\mathbb{Q}) \cap H^{p,p}(A), \qquad Z^{2p}(A) = \mathrm{im}\big(\mathrm{CH}^p(A)_{\mathbb{Q}} \xrightarrow{\mathrm{cl}} H^{2p}(A,\mathbb{Q})\big).$$

The source [1] claims equality for every CM A and every $0 \le p \le g$, including products and powers. A CM variety here may have a commutative semisimple endomorphism algebra of dimension $2g$; simplicity and a primitive CM type are not assumed. This report proves finite deductions relevant to that claim. Reading a manuscript, compiling this report, and passing finite tests do not independently validate the claimed equality or its arithmetic consequences.

All twelve cached TeX files were literally read in bounded displays, including the complete arithmetic extraction and consequence sections. The bibliography was read only in selected intervals. The accompanying read ledger records each display, its unread complement, and pre/post hashes. Private originals were read locally in selected passages only: the Hodge report [5], the original stable-frame report [6], and the original compatible-complex report [7]. No whole-corpus reading is claimed. Parent checks and snapshot authentication are separate and are not author-run tests.

1.1 What the source actually needs

Source module	Logical role and retained boundary
Introduction	States the global claim and distinguishes absolute Hodge classes, quadratic period relations, and actual algebraic cycles.
02: tensors	All-conjugate trace descent; weight-one maps; CM sign modules; reduction to four-factor switches; auxiliary-product contraction. Our switch-count improvement applies here.
03: surface	A nonzero fourfold cup period on one smooth projective surface gives a functional. Four divisor kernels convert it to a vector on the required line. The surface and rational maps are essential.
04: theta	Positive Lagrangians, completed Taylor tensors, rational Weil covariance, unitary splitting, sparse-gradient spanning, double alternation, and fixed-total-input continuity. Our tail estimate applies to its sparse limit, not to the arithmetic rationality step.
05a: moduli	Full polarized, acted-on, marked tuple space; projectivity; ordinary points on every component; integral Hecke action on the product of Albanese varieties. These global constructions remain imported.
05b: Frobenius	Counts ordinary isogeny neighbors with multiplicities and full markings, then proves a cubic identity on Albanese differences. Its specialization and density premises are not replaced by finite tests.
05b: Satake	Inverse spherical action; radial dilation; rational scalar covariance; conjugate valuations and finite-order central operators. All normalization choices remain part of the imported argument.
05c: CM types	Generalized Hecke idempotents preserve the filtration; Weil weight selects the Frobenius root; weak admissibility determines Hodge types of every scalar conjugate; trace descent gives rational maps.
06: assembly	Finite endpoint wedge expansions are moved to one projective cover; nonzero total period selects a nonzero term; CM maps feed the surface criterion.
07: consequences	Effective CM realizations and correspondence supports; generalized Hodge, Tate, standard-conjecture and specialization claims also depend on the global CM theorem and cited external inputs.

No substantive mathematical correction to an inspected source step is established here. The local bounds below strengthen the source's finite implementation, not its theorem statement. Retained details include separate repeated embedding slots, scalar conjugation on the same variety, the positive Lagrangian rather than its dual, the $1/k!$ Taylor normalization, and common levels chosen only after finite input selection.

1.2 Primary literature refreshed on October 8

Gao–Ullmo's Theorem 1.1 [2] adjoins an auxiliary CM factor and generates Hodge-induced holomorphic period relations by explicit quadratic ones. Its Theorems 7.1–7.2 describe the relevant balanced four-label relations and generation for generalized anti-Weyl varieties. These are Hodge relations, not an automatic construction of algebraic cycles. The publisher dates online publication

to September 19, 2025; the source's 2026 citation is the volume year, not an error. We inspected those main statements and definitions, not all proofs.

Markman's later account [3] states algebraicity of Weil classes on abelian fourfolds of Weil type and sixfolds of split Weil type for imaginary quadratic multiplication (Theorem 1.2), and the Hodge conjecture in dimension at most five (Corollary 1.3). Thus the source introduction's mention of dimension four is not the strongest current statement in that account. This is a literature update, not a contradiction. The more general real-multiplication strategy [4] explicitly leaves semiregularity unaddressed in its advertised higher-dimensional example. The inspected statements do not supply the source's full CM theorem.

2 CM signs, exact tensors, and scalar descent

Fix a Galois CM field $E \subset \overline{\mathbb{Q}}$, $F = E^+$, $s = [F : \mathbb{Q}]$, $G = \mathrm{Gal}(E/\mathbb{Q})$, and its central conjugation c. Let $\Sigma \subset G$ contain one representative of each pair $\{\tau, c\tau\}$. An odd sign function is $v : G \to \{+1, -1\}$ with $v(c\tau) = -v(\tau)$. The rational space $U(v) = E$ has embedding lines $U(v)_\lambda$, of type $(1, 0)$ for $v(\lambda) = 1$ and $(0, 1)$ otherwise. The source realizes it as $H^1(B(v), \mathbb{Q})$ for a polarized abelian variety of dimension s. Its polarization uses an imaginary $\zeta \in E$ and $\mathrm{Tr}_{E/\mathbb{Q}}(\zeta x c(y))$, with signs chosen for positivity. Neither principal polarization nor a canonical lattice is required.

For two ordered lists $a = (a_1, \dots, a_p)$ and $b = (b_1, \dots, b_p)$, put $A_a = \prod_i B(a_i)$ and

$$\mathscr{L}_{a,b} = \bigotimes_{i=1}^{p} U(a_i)_1 \otimes \bigotimes_{i=1}^{p} U(b_i)_c \subset H^{2p}(A_a \times A_b, \overline{\mathbb{Q}}). \tag{1}$$

All slots are distinct, even when their types agree. Under scalar conjugation restricting to $g \in G$, the labels $1, c$ become g, gc. Consequently all conjugates of this line have type (p, p) precisely when

$$\sum_{i=1}^{p} a_i(g) = \sum_{i=1}^{p} b_i(g) \quad (g \in G). \tag{2}$$

Indeed its first Hodge index is $p + \frac{1}{2}\sum_i (a_i(g) - b_i(g))$. This is an exact finite balance test. It does not itself prove algebraicity.

Lemma 2.1 (Trace descent, retained from the source)**.** *Let T be a finite rational space and $P \subset T \otimes \mathbb{C}$ a complex linear subspace. If $t \in T \otimes \overline{\mathbb{Q}}$ and every scalar conjugate of t lies in P, then $t \in (T \cap P) \otimes \overline{\mathbb{Q}}$.*

Proof. Take a finite Galois coefficient field K and a basis e_j with trace dual basis $e_j^\vee$. Write $t = \sum_j t_j e_j$, $t_j \in T$. Then $t_j = \mathrm{Tr}_{K/\mathbb{Q}}(e_j^\vee t)$ is a linear combination of conjugates of t, hence lies in P. This proves the claim. The same argument applies to the subspace of maps of a fixed Hodge type. □

If an algebraic-coefficient cycle span contains a rational class, it contains that class over $\mathbb{Q}$: extend a rational basis of the cycle span to a rational basis of the ambient cohomology. Thus final scalar descent is linear algebra on an already established cycle span. It is not a procedure for descending a numerically guessed geometric cycle.

Definition 2.2 (Geometric switch input)**.** For each odd quadruple u, v, u', v' with $u + v = u' + v'$, assume the line $U(u)_1 \otimes U(v)_1 \otimes U(u')_c \otimes U(v')_c$ has a nonzero generator in the algebraic cycle span with $\overline{\mathbb{Q}}$ coefficients on the specified auxiliary product. This is the source's four-factor theorem, imported as a hypothesis whenever used below.

The hypothesis includes actual cycle algebraicity, not just a Hodge tensor, a relation among periods, or a positive numerical rank.

3 Batched switches with an exact construction budget

Represent a, b on Σ by sign matrices $X, Y \in \{\pm 1\}^{p\times s}$. Equation (2) says their column sums agree. For column k, let m_k be half the number of mismatching entries, and set $M = \sum_k m_k$. Then $m_k \le \lfloor p/2 \rfloor$. A batched switch selects two rows and exchanges their entries in an arbitrary set of columns, also exchanging the conjugate coordinates with opposite signs. The old and new two-row sums agree on all of G. Thus one such operation requires at most one geometric switch input, regardless of the number of columns exchanged.

Theorem 3.1 (Monotone batched reduction)**.** *For all $p, s \ge 1$ and balanced X, Y, there is a sequence of nonempty batched switches from X to Y of length*

$$N \le \min\{M, \tbinom{p}{2}\} \le \min\{s\lfloor p/2 \rfloor, \tbinom{p}{2}\}. \tag{3}$$

Every changed entry is changed exactly once, from an incorrect to a correct value. At most $p + 2N$ distinct row types need occur along the sequence, counted before possible identifications.

Proof. Start with the current matrix $Z = X$. Process pairs (i, j) in lexicographic order, $1 \le i < j \le p$. At pair (i, j) exchange simultaneously the columns in

$$D_{ij} = \{k : Z_{ik} \ne Y_{ik},\ Z_{jk} \ne Y_{jk},\ Z_{jk} = Y_{ik}\}.$$

Binary signs imply $Z_{ik} = Y_{jk}$ in these columns, so both changed entries become correct. No correct entry is ever changed. At the start of row i, earlier rows are correct. If entry (i, k) is still wrong, balance among the remaining rows gives a later wrong entry with the opposite desired sign. Such an entry remains available until matched: earlier operations on row i in other columns cannot affect it. Processing the later rows therefore fixes (i, k). Induction fixes all rows; the last row also follows from column balance.

There are $\binom{p}{2}$ row pairs, each used once. Every nonempty operation removes at least two mismatches, while each individual column exchange removes exactly two. There are exactly M such column exchanges, proving both bounds. Each operation introduces at most two new row types. □

The source's single-column construction makes M elementary corrections. Grouping gives a potentially strict saving without changing its class of admissible four-factor relations. This is a construction-count bound, not a bound on degrees, heights, field discriminants, or the cost of constructing one surface. It can be unhelpful for small s; taking the minimum is essential.

Example 3.2 (An unbounded saving at fixed codimension)**.** Take $p = 2$, X with first row all $+$ and second row all $-$, and Y obtained by exchanging any $t > 0$ columns. The source's one-column path has t steps; the batched path has one. Keep one unchanged positive distinguished coordinate when desired, and at least one exchanged coordinate, so the four types need not be global permutations of the same pair. In particular this is not merely a renaming of rows. The four-factor geometric input is still needed unless the resulting tensor separately factors into known divisor pairs.

3.1 Optimal monotone batching as a finite certificate

For column k let $P_k = \{i : X_{ik} = +, Y_{ik} = -\}$ and $Q_k = \{j : X_{jk} = -, Y_{jk} = +\}$. They have equal size m_k. A perfect matching between them specifies all corrections in that column. Use binary variables $u_{e,k}$ for allowed unordered pairs $e = \{i, j\}$ with one endpoint in each set, and z_e for all row pairs.

Proposition 3.3 (Exact support formulation)**.** *The minimum number of operations among monotone batched paths equals*

$$\min \sum_e z_e, \quad \sum_{e \ni i} u_{e,k} = 1 \ (i \in P_k \cup Q_k), \qquad u_{e,k} \le z_e, \qquad u_{e,k}, z_e \in \{0, 1\}. \tag{4}$$

In particular every feasible matching collection gives a verifiable path, and every such path uses at least

$$\max\{\max_k m_k, \lceil r/2 \rceil\}$$

operations, where r is the number of initially incorrect rows.

Proof. In a monotone path every wrong entry is corrected once by an oppositely wrong entry, giving the stated perfect matchings. Repeated use of one row pair can be grouped. Conversely, group a feasible matching collection by its row pairs and perform these groups in any order. In each column the matching edges are disjoint, so no entry is touched twice and all corrections are valid. Thus the union of the edge supports is exactly the number of nonempty groups. A switch corrects at most one mismatch pair per column and touches only two rows, proving the lower bounds. □

This formulation asserts no optimality among paths that deliberately break correct entries, and supplies no polynomial-time optimizer. A feasible certificate is checked in $O(ps + p^2 s)$ elementary operations; optimality requires an independent lower bound or exact finite optimization.

3.2 Why the shorter path really composes algebraically

For every transition, tensor its supplied four-factor class with the $p - 2$ unchanged two-factor classes. The latter are algebraic by Lemma 2.1 and Lefschetz $(1, 1)$. This produces an algebraic generator of (1) on the old and new products.

Proposition 3.4 (Normalized composition)**.** *Under Definition 2.2, the path of Theorem 3.1 gives an algebraic generator of $\mathscr{L}_{a,b}$ using at most N four-factor inputs and $\max\{N - 1, 0\}$ middle-product contractions. For $N = 0$, ordinary two-factor classes suffice. For a contraction through an ordered middle list d, the exact multiplier is*

$$\kappa_d = (-1)^{p(p-1)/2} \prod_{i=1}^{p} \int_{B(d_i)} \beta_i \wedge \gamma_i \wedge \ell_i^{s-1} \in \overline{\mathbb{Q}}^{\times}, \tag{5}$$

where $\beta_i \in U(d_i)_c$, $\gamma_i \in U(d_i)_1$ are the nonzero algebraic Betti vectors in the two generators and ℓ_i is any ample rational divisor class.

Proof. Pull the adjacent classes to $A_a \times A_d \times A_b$, cup with $\prod_i \ell_i^{s-1}$, and push to $A_a \times A_b$. Moving the ordered middle factors $\beta_1 \cdots \beta_p \gamma_1 \cdots \gamma_p$ into paired order causes $p(p-1)/2$ odd-degree interchanges. The ample powers have even degree. Integrating the middle product gives (5). Its factors are nonzero by Hodge–Riemann positivity: the two middle lines are complex conjugates and are one-dimensional. The factors are algebraic because all Betti vectors have algebraic coefficients and the integration

pairing and ample classes are rational. Codimension is $p+p+p(s-1)-ps=p$. All operations are operations on actual algebraic cycle spans, so division by κ_d is legitimate over $\overline{\mathbb{Q}}$. Induct on the number of transitions. □

For unit-normalized transition generators the accumulated scalar is the product of the successive κ_d. It must be recorded or canceled exactly, not discarded as "nonzero up to numerical error." If intervals are used to establish nonvanishing, each scalar enclosure must exclude zero; this is not an integral saturation assertion. Fewer contractions reduce the number of required certificates, but no universal improvement in their conditioning follows. The source's final line transfers and diagonal pullback to A remain necessary. Distinct original exterior slots guarantee that last cup product is nonzero. Nothing here says that the Hodge ring on a fixed power is generated by divisor and codimension-two classes on that same power.

4 A quantitative sparse theta-gradient certificate

The source constructs one-forms from a multiplicity line over a totally real field F of degree s. Fix one period point $Z=(Z_\tau)$, with $Z_\tau\in\mathfrak{H}_3$ and $\operatorname{Im} Z_\tau\succeq\lambda I$ for a certified $\lambda>0$. Use the Euclidean norm in the Minkowski position space $\mathbb{R}^d$, $d=3s$. Let L be its full lattice $\mathcal{O}_F^3$, with certified minimum nonzero vector length at least $\rho>0$. Fix $0\neq x_0\in L$ and the finite input supported on $x_0+k\widehat{\mathcal{O}}_F^3$. At the point under consideration, let P be the graph-frame, half-determinant and cotangent projection into a fixed Hermitian two-dimensional cotangent space, with operator norm at most K. These choices, including the constant-line trivialization, are fixed. The projected first Taylor tensor and its single-term limit are

$$\begin{aligned} g_k &= 2\pi i P \sum_{x\in x_0+kL} x\exp(\pi i\sum_\tau x_\tau^t Z_\tau x_\tau), \\ v_0 &= 2\pi i P x_0\exp(\pi i\sum_\tau x_{0,\tau}^t Z_\tau x_{0,\tau}). \end{aligned} \tag{6}$$

No Taylor factorial has been omitted: the linear coefficient is exactly the first derivative. The quadratic completion in the source contributes no linear term.

Theorem 4.1 (Explicit lattice-shell tail)**.** *Assume $k\rho\geq 2\|x_0\|$ and put*

$$a=\frac{\pi\lambda k^2\rho^2}{4},\qquad r_a=\frac{3}{2}\left(\frac{7}{5}\right)^d e^{-3a}.$$

Then

$$\|g_k-v_0\|\leq 3\pi Kk\rho\sum_{l=1}^{\infty}(l+1)(2l+3)^d e^{-al^2}. \tag{7}$$

If $r_a<1$, a closed upper bound is

$$\|g_k-v_0\|\leq\frac{6\pi Kk\rho\,5^d e^{-a}}{1-r_a}=:\varepsilon_k. \tag{8}$$

The bounds are uniform on a declared compact family with the same λ,K,ρ,x_0. They assert no bound on the congruence level or arithmetic height of the input.

Proof. For $m\in L\setminus\{0\}$,

$$\tfrac{1}{2}k\|m\|\leq\|x_0+km\|\leq\tfrac{3}{2}k\|m\|.$$

The modulus of the exponential in (6) is at most $e^{-\pi\lambda\|x_0+km\|^2}$, so the summand norm after projection is at most $3\pi Kk\,\|m\|\,e^{-\pi\lambda k^2\|m\|^2/4}$. In the shell $l\rho \le \|m\| < (l+1)\rho$, this is bounded by $3\pi Kk(l+1)\rho e^{-al^2}$. Balls of radius $\rho/2$ centered at lattice points are disjoint up to boundaries. Those with centers of norm less than $(l+1)\rho$ fit inside the ball of radius $(l+3/2)\rho$. Comparing volumes bounds their number by $(2l+3)^d$. This proves (7), without assuming an orthogonal lattice.

For $u_l = (l+1)(2l+3)^d e^{-al^2}$,

$$\frac{u_{l+1}}{u_l} = \frac{l+2}{l+1}\left(\frac{2l+5}{2l+3}\right)^d e^{-a(2l+1)} \le r_a \quad (l \ge 1).$$

Sum the geometric majorant with $u_1 = 2\cdot 5^d e^{-a}$. Uniformity uses only the displayed shared constants. □

The estimate deliberately overcounts the lattice points. For small a, sum the first $L-1$ shell bounds and bound the remaining tail from L using the ratio

$$r_{a,L} = \frac{L+2}{L+1}\left(\frac{2L+5}{2L+3}\right)^d e^{-a(2L+1)} < 1.$$

All three factors in the ratio decrease with L, so the same proof works. Validated transcendental enclosures, not floating-point exponentials alone, are required in a numerical application.

Corollary 4.2 (Finite-input wedge margin)**.** *For two multiplicity lines choose centers and fixed projections as above, with limiting cotangent vectors v_1, v_2. In an orthonormal cotangent frame let $\gamma = |\det(v_1, v_2)| > 0$. If the two finite-input errors obey*

$$\varepsilon_1\,\|v_2\| + \varepsilon_2\,\|v_1\| + \varepsilon_1\varepsilon_2 < \gamma, \tag{9}$$

the actual one-forms have a nonzero wedge at the selected point.

Proof. Expand $\det(v_1+e_1, v_2+e_2)$ and use $|\det(u,w)| \le \|u\|\,\|w\|$. Its difference from the limiting determinant has precisely the upper bound in (9). □

The source proves existence of suitable centers because the embedded lattice spans the real position space, its complexification spans the whole first-tensor fiber, and each cotangent projection is surjective. This supplies independent limiting vectors from the two families. Our bound then gives a finite stopping condition for the source's sparse limit, conditional on certified λ, ρ, K, γ. It does not provide those constants for a particular CM field in this report. Choose the finite inputs first; choose a common principal stabilizer afterward. There need not be a common level for all $k \to \infty$.

Example 4.3 (A manufactured certificate scale)**.** As a bound example only, take $d = 6$, $\lambda = \rho = K = 1$, $\|x_0\| \le 1$, and $k = 10$. Then $a = 25\pi > 75$. Using $\pi < 4$, $e > 2$ gives $r_a < \frac{3}{2}(7/5)^6 2^{-225} < 1/2$ and

$$\varepsilon_k < 480\cdot 5^6\,2^{-75} < 1/1000.$$

Two orthonormal limiting vectors would therefore satisfy (9), since $2/1000 + 1/10^6 < 1$. This checks the stopping rule's scale, not the existence of a CM surface with these normalized data. Its constants are manufactured, not measured periods and not a numerical experiment on an abelian variety.

5 Mixed periods and the geometry gate

On a fixed compact complex surface, top forms carry the positive Hermitian pairing $\langle \xi, \eta \rangle = \int \xi \wedge \overline{\eta}$ in complex dimension two. With the complex orientation, $dz_1 \wedge dz_2 \wedge d\bar{z}_1 \wedge d\bar{z}_2$ is positive. In particular

$$|\langle \xi, \xi' \rangle| \geq \|\xi\|_2 \left(\|\xi\|_2 - \|\xi' - \xi\|_2 \right). \tag{10}$$

This is a quantitative form of the source's fixed-total-input continuity step, not a substitute for establishing that continuity on one quotient. Intermediate parameter forms need only be smooth of type $(2,0)$; holomorphy is asserted at the rational endpoints.

Proposition 5.1 (A complete scalar period budget)**.** *Suppose* $\left\| \xi - \widehat{\xi} \right\|_2 \leq a$ *and* $\|\eta - \widehat{\eta}\|_2 \leq b$. *Suppose a computed scalar* $\widehat{I}$ *approximates* $\langle \widehat{\xi}, \widehat{\eta} \rangle$ *with certified error at most* e_Q. *Then*

$$|\langle \xi, \eta \rangle - \widehat{I}| \leq e_Q + a \|\widehat{\eta}\|_2 + b \left\| \widehat{\xi} \right\|_2 + ab =: e_I. \tag{11}$$

If $|\widehat{I}| > e_I$, *the true period is nonzero. If, moreover,* $\xi = \sum_i c_i \omega_{1i} \wedge \omega_{2i}$ *and* $\eta = \sum_j d_j \omega_{3j} \wedge \omega_{4j}$ *are finite expansions, then some individual mixed period has modulus at least*

$$\frac{|\widehat{I}| - e_I}{(\sum_i |c_i|)(\sum_j |d_j|)}. \tag{12}$$

Proof. Expand both approximation errors and apply Cauchy–Schwarz to the three error pairings. Expand the exact period as $\sum_{i,j} c_i \overline{d_j} P_{ij}$ and apply the triangle inequality. The nonzero total ensures the denominator in (12) is positive. The same estimates apply with certified upper bounds for it. □

To select a specific term algorithmically, compute individual interval enclosures until one excludes zero. Existence follows from the bound; termination also requires convergent enclosures for those fixed periods. For an unramified finite cover of degree h, the integral and squared norms scale by h, and norms by $\sqrt{h}$. A normalized scalar budget must respect these factors. No cusp or ramification term is suppressed.

5.1 The exact surface-to-cycle implication, with its premises

Suppose S is smooth connected projective, and actual rational Hodge maps $h_i : U(v_i) \to H^1(S, \mathbb{Q})$ are supplied. Let α_i be nonzero algebraic Betti vectors with labels $(1,1,c,c)$. If their cup period is nonzero, weight-one functoriality gives morphisms $f_i : S \to B(v_i)$ after clearing positive integers m_i, with $f_i^* = m_i h_i$. For $B = \prod_{i=1}^4 B(v_i)$, $n = 4s$, set $Z = \mathrm{cl}(f_*[S]) \in H^{2n-4}(B, \mathbb{Q})$. The period $k = \int_B Z \cup (\bigotimes_i \alpha_i)$ is in $\overline{\mathbb{Q}}^\times$.

Choose opposite-label algebraic Betti vectors β_i on copied factors. Each $D_i = \alpha_i \otimes \beta_i$ is an all-conjugate $(1,1)$ class, thus in the algebraic divisor span by trace descent and Lefschetz. On $B \times B'$ let $D = \prod_i D_i$. Reordering four old and four new degree-one factors has sign $(-1)^6 = 1$. Therefore

$$(p')_*(p^* Z \cup D) = k \bigotimes_i \beta_i. \tag{13}$$

Codimension is $(n-2) + 4 - n = 2$. This is an algebraic vector, not merely the functional furnished by Z. Dividing by k and applying scalar conjugation gives the desired switch line. If the one-forms are only finite complex spans of rational Hodge images, expand the nonzero period first to select one nonzero term. The source proves precisely this criterion; we retain it rather than claim it as new.

The missing premise is not a precision parameter. A certified nonzero wedge, a period relation, and full numerical rank do not supply S, the rational h_i, or the morphisms f_i. In the source these come from its separate moduli and arithmetic extraction arguments. Our estimates can certify nonvanishing once the geometric objects and comparison maps are established; they cannot create them.

6 Projectors and rational-span certificates

We record an exact implementation of the finite projector interface. These classical identities and the private certificate principles are supporting tools, not the scored novelty of this report.

6.1 Algebraic CM projectors with the correct degree

On a dimension-s abelian variety, choose an integer $n \geq 2$ and put

$$\pi_1 = \prod_{j=0,\ j\neq 1}^{2s} \frac{[n]^* - n^j I}{n - n^j}. \tag{14}$$

Multiplication by n acts on H^j as n^j, so this is the degree-one projector. It is induced by a rational algebraic correspondence, being a polynomial in pullbacks and the diagonal. Take a primitive element θ of the CM field acting on H^1, and an integer $m > 0$ making $u = m\theta$ an actual endomorphism. The rational correspondence operator $T = m^{-1}u^*$ has the desired action on H^1. For an embedding λ,

$$e_\lambda = \pi_1 \prod_{\nu\neq\lambda} \frac{T - \nu(\theta)I}{\lambda(\theta) - \nu(\theta)}. \tag{15}$$

The denominators are nonzero by separability. On H^1, interpolation gives mutually orthogonal idempotents summing to the identity, while π_1 kills other degrees. One must not interpret $m^{-1}u^*$ as the pullback of the rational endomorphism θ in all degrees.

External products select the embedding slots of an auxiliary product. For a complete scalar-Galois orbit $\mathcal{O}$ of these label tuples, the sum $e_\mathcal{O}$ is a rational polynomial combination of the underlying rational correspondences: its coefficients are Galois fixed. It is an idempotent on the specified cohomology and preserves the algebraic cycle span. An individual e_λ need not be rational; an orbit sum need not be a single line or a full Hodge space. Exact orbit membership, the Hodge balance test, and an exterior multiplicity census remain separate checks. These projectors control known cycles, not their existence.

6.2 Generalized Hecke blocks and a filtration warning

The source uses product idempotents in a finite commutative algebra, not an unjustified simultaneous eigenbasis. If one separating operator has annihilating polynomial $\prod_i (t-\lambda_i)^{h_i}$, its generalized projector is obtained by Chinese remaindering: $p_i \equiv 1$ modulo $(t-\lambda_i)^{h_i}$ and $p_i \equiv 0$ modulo all the other factors. Equivalently invert the other factors modulo $(t-\lambda_i)^{h_i}$. Evaluation proves $p_i^2 = p_i$, $p_i p_j = 0$, and $\sum p_i = I$. The simple Lagrange formula without multiplicities would generally fail in the presence of Jordan blocks. A separating operator and the necessary annihilating polynomial must first be known exactly.

For the source's degree-one slope argument, the projector and its complement preserve the Hodge filtration and commute with Frobenius. Weak admissibility gives $t_H \leq t_N$ on both summands;

equality on the whole forces equality on each. A pure slope $t \in \{0, 1\}$ then gives $\dim F^1 = t \dim D$ because the only filtration indices are $0, 1$. Without filtration preservation this implication fails. On $D = \mathbb{Q}_q e_0 \oplus \mathbb{Q}_q e_1$ with $\varphi = \mathrm{diag}(1, q)$ and $F^1 = \langle e_0 + e_1 \rangle$, weak admissibility holds but the slope-one line has induced $F^1 = 0$. This is already an explicit warning in the source. We do not infer Hodge type from a numerical spectral cluster.

6.3 A finite certificate after genuine cycle construction

Let $V_{\mathbb{Q}}$ be an exactly identified rational subspace of $H^{2p}(A, \mathbb{Q})$ of dimension r. Suppose actual cycles $Z_1, \dots, Z_m$ have classes in $V_{\mathbb{Q}}$, perhaps after applying a known rational algebraic projector. Fix a rational basis and injective comparison tests giving a nonsingular matrix B and the cycle pairing matrix P. The coordinate matrix $C = B^{-1}P$ is rational because of those exact premises, not because of its numerical appearance. With certified norm errors $\left\|B - \widehat{B}\right\| \le e_B$, $\left\|P - \widehat{P}\right\| \le e_P$, $\left\|\widehat{P} - \widehat{B}\widehat{C}\right\| \le e_S$, and $s_B \le \sigma_{\min}(\widehat{B})$, $s_B > e_B$, one has

$$\left\|C - \widehat{C}\right\| \le \frac{e_P + e_B \left\|\widehat{C}\right\| + e_S}{s_B - e_B} =: e_C. \tag{16}$$

Indeed subtract $B\widehat{C}$ from P and use $\|B^{-1}\| \le (s_B - e_B)^{-1}$. A certified lower bound for $\sigma_r(\widehat{C})$ exceeding e_C proves full row rank. Rational elimination then proves spanning over $\mathbb{Q}$, not just over $\mathbb{C}$. This is the original private finite-span criterion [5], with frame errors accounted for as in [6]; it is not a new Hodge theorem.

Likewise, suppose a proposed equality $v = \alpha - Cb$ has exact coordinates in $(DQ)^{-1}\mathbb{Z}^r$, and a test map T on the whole comparison space satisfies $\sigma_{\min}(T) \ge s_0 > 0$. Then

$$\|Tv\| < s_0/(DQ) \quad \Longrightarrow \quad v = 0. \tag{17}$$

A nonzero lattice vector has norm at least $1/(DQ)$, proving the implication. Denominator bounds and injectivity must precede this test. No unknown saturation index may be silently rounded away. The transformation of a finite cochain quotient under an invertible coordinate map, or under a chain retraction with a homotopy, preserves cohomology [7]; it does not turn a rational cochain into an algebraic cycle. We use this original distinction, not a derivative summary as evidence for it.

7 Boundaries, tests, and assessment

7.1 Independent finite checks

The accompanying author script uses integers, rational numbers, and symbolic polynomials. It exhaustively checks selected small balanced sign matrices and tests larger deterministic instances. For each path it checks unchanged column sums, exactly-once mismatch corrections, pair uniqueness, endpoint agreement, and both switch-count bounds. It compares the monotone matching optimum with a direct finite state search in small cases, checks the polarization regrouping sign by inversions, and checks the scalar contraction codimension. Further checks cover the shell ratio and finite partial-sum majorants with rational $q = e^{-a}$, the manufactured stopping margin, CM and generalized-block interpolation, and strict period/lattice thresholds. The completed suite passes 353,827 exact assertions in 25,542 cases, including 22,734 exhaustive paths, 500 seeded paths, 160 comparisons of matching and state-search optima, and 216 shell-majorant cases. The frozen JSON gives all category counts and the script hash. These are manufactured finite tests of local formulas, not

measured CM periods, surface constructions, arithmetic specialization tests, or independent proofs of the source theorem. No parent's tests are included.

7.2 Advantages and costs

Advantages. Batching removes the embedding count from one upper bound on four-factor invocations. Matching support makes the optimization target precise and auditable. The contraction formula transports that saving to the actual conditional cycle construction. Sparse-gradient tails give a finite analytic stopping rule in place of an unspecified limit. Explicit projection, comparison, and denominator premises prevent numerical rank from being mistaken for algebraicity.

Costs. Large-degree CM fields still make individual geometric switch inputs expensive; the report bounds neither those costs nor final cycle degree. The shell estimate may be very conservative in high real dimension. Minimum lattice separation, period eigenvalue margins, and frame norms need independent certification. Fewer factors need not mean better conditioning. Exact rational orbit projectors may have large denominators. The monotone optimum is not asserted to be a globally shortest unrestricted path.

7.3 Feasible extensions and research priorities

An implementation can cache identical row types and supplied switch cycles, solve the matching-support problem on a fixed input, and compare actual contraction budgets rather than just operation counts. Any such optimization must retain intermediate products and maps. An anisotropic lattice ellipsoid count could improve (7), but would require certified shape information rather than a minimum-distance bound alone. Derivative estimates for the fixed-total theta family could replace continuity by an explicit rational-approximation radius; the finite input, quotient, and smooth trivializations must remain fixed throughout.

The highest-priority independent geometric work is validation of the source's projective tuple moduli, all-component ordinary specialization, Hecke correspondence multiplicities, and rational-source extraction with every scalar conjugate. No theorem here weakens those requirements. Specialization to characteristic p, generalized Hodge supports, and Tate or standard-conjecture consequences are not outputs of a finite period calculation. Those remain imported consequences conditional on their own stated global inputs.

7.4 Contribution judgment

Applicability remains L3. The demonstrated contribution is C3: a source-specific reduction in the number of geometric four-factor invocations, with an exact monotone optimization model and preserved contractions, plus a quantitative sparse-theta stopping certificate. The matching idea and Cauchy–Schwarz/packing estimates use classical mathematics; no priority claim over all literature is made. Trace descent, Lagrange/Chinese-remainder projectors, and the private rank/lattice criteria are credited as existing tools, not relabeled contributions. Neither a new family of cycles nor an independent proof of the rational Hodge conjecture for CM abelian varieties has been established here.

References

[1] OpenAI, *The rational Hodge conjecture for CM abelian varieties*, September 30, 2026, cached manuscript at pinned repository commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public source directory.

[2] Z. Gao and E. Ullmo, *Hodge cycles and quadratic relations between holomorphic periods on CM abelian varieties*, J. Inst. Math. Jussieu 25 (2026), 215–249; online September 19, 2025. doi:10.1017/S1474748025101291. Main statements 1.1, 7.1–7.2 and definition 1.3 inspected.

[3] E. Markman, *Secant sheaves and Weil classes on abelian varieties*, arXiv:2509.23403v2. Theorem 1.2, Corollary 1.3, and surrounding introduction inspected; not a complete proof audit.

[4] E. Markman, *Secant sheaves on abelian n-folds with real multiplication and Weil classes on abelian $2n$-folds with complex multiplication*, arXiv:2509.23079v1. Abstract's semiregularity limitation inspected; not all proofs.

[5] M. Subbarao, original confidential Hodge manuscript, `res-math/TR-item-011-astra.tex`, locally read lines 115–308 and 471–695. Period, perturbation, closure, lattice-gap, and symmetry proofs; no private upload or public link.

[6] M. Subbarao, original confidential TR248, *Stable Adaptive Ram Frames*, locally read lines 349–405 and 1030–1120. Physical frame margins, analytic tails, and exact certificates.

[7] M. Subbarao, original confidential TR250, *Compatible Variational Ram Panels*, locally read lines 207–238 and 724–769. Coordinate quotients and topology-preserving compression.